Bounded Successor Distance in the Reversible Square-Multiplication Graph

UNNS-U1-BND – version 1.0 (v001)

UNNS Research Collective

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Abstract

Let G be the graph on the positive integers in which kN is joined to $k(N+1)^2$ for all $k, N \geq 1$, and let d denote its graph distance. We prove that consecutive integers are always close: $2 \leq \mathrm{d}(n, n+1) \leq 6$ for every $n \geq 1$, and the value 6 is attained, so that $\sup_n \mathrm{d}(n, n+1) = 6$. The upper bound is obtained from nine explicit path identities, one for each residue class of n modulo 4, 8 or 24. In each class the path from n to $n+1$ has a fixed number of steps and its labels are linear polynomials in a class parameter; every step is certified by an integral, positive cofactor, so legality holds for all members of the class at once. The lower bound is an exhaustive meeting-in-the-middle computation at $n = 163$, reproduced by two independent search programs. Consequently the total distance along $1, 2, \ldots, k+1$ is at most $6k$.

1 Introduction

Provenance. The graph G is the undirected closure of the arithmetic transformation $m \mapsto m(N+1)^2/N$ $(N \mid m)$ of the SELECT-II grammar of the UNNS Substrate: its forward edges are the legal transitions and its inverse edges are their inverses. Forward-only properties of this transformation (its exact one-step image, orbit criterion and label-realizability criterion) are the subject of research note U0-001 [1]. The connectivity of G is established in the preceding note [2], which is kept as a provisional baseline. The present release, UNNS-U1-BND, is independent of both: it leaves them, and the frozen reference collection [3, 4], unchanged. Its question is quantitative. Given that G is connected, how far apart can *consecutive* integers be?

Result.

Theorem 1.1 (Bounded successor distance)**.** *For every integer* $n \geq 1$, $2 \leq \mathrm{d}(n, n+1) \leq 6$. *Moreover* $\mathrm{d}(163, 164) = 6$. *Hence*

$$C_* := \sup_{n \geq 1} \mathrm{d}(n, n+1) = 6.$$

Mechanism. The identities share one idea: a forward step produces a number carrying a large square factor built from n or $n+1$, and the surplus is then traded for the target by inverse steps whose labels are linear in a class parameter. Which linear forms are integral depends on n modulo 8 and, in the hardest classes, modulo 3; this is why the constant becomes 6 only after splitting modulo 24 (Section 3).

Organization. Section 2 fixes notation. Section 3 proves the upper bound; Section 4 the sharpness. Section 5 reports the exact distances computed. Section 6 separates what is proved from what is only computed. Section 8 lists open questions. Appendix A contains the complete step-by-step certificates.

2 Preliminaries

Definition 2.1. G has vertex set $\mathbb{N}_{>0}$ and an edge $\{kN,\ k(N+1)^2\}$ for each pair $k, N \geq 1$. A *forward step with label* N replaces $x = kN$ by $k(N+1)^2$; an *inverse step with label* N replaces $x = k(N+1)^2$ by kN. The number k is the *cofactor* of the step. d is the shortest-path distance.

A forward step with label N is legal exactly when $N \mid x$; an inverse step with label N exactly when $(N+1)^2 \mid x$. Each vertex has finitely many neighbours (forward labels are divisors of x, inverse labels satisfy $(N+1)^2 \mid x$), so G is locally finite and balls of finite radius are finite, computable sets.

Lemma 2.2 (Edge criterion)**.** *For $x < y$, $x \sim y$ if and only if $y = x(N+1)^2/N$ for some integer $N \geq 1$ with $N \mid x$, and then N is a positive root of $N^2 + (2 - y/x)N + 1 = 0$.*

Proof. Immediate from the definition; $y/x = (N+1)^2/N$ rearranges to the stated quadratic. □

Lemma 2.3 (Consecutive integers are never adjacent)**.** $\mathrm{d}(n, n+1) \geq 2$ *for all* $n \geq 1$.

Proof. An edge multiplies by $(N+1)^2/N = N + 2 + 1/N \geq 4$, whereas $(n+1)/n \leq 2$. □

3 The upper bound: nine path identities

Table 1 lists the identities. In it $+N$ denotes a forward step and $-N$ an inverse step with label N; t is a nonnegative integer parameter with the stated lower limit.

class	n	steps	labels (forward $+N$, inverse $-N$)
A	$n = 4t+1\ (t \geq 1)$	4	$+(4t+1)$, $-2t$, $+t$, $-(2t+1)$
B	$n = 8t\ (t \geq 1)$	5	$+2$, $+2$, -8, $+4t$, $-(8t+1)$
C	$n = 8t+2\ (t \geq 0)$	5	$+2$, $+(4t+1)$, $+2$, -8, $-(8t+3)$
D	$n = 8t+4\ (t \geq 0)$	5	$+2$, $+2$, -8, $+(2\,(2t+1))$, $-(8t+5)$
E	$n = 8t+6\ (t \geq 0)$	5	$+2$, $+(4t+3)$, $+2$, -8, $-(8t+7)$
F	$n = 24t+3\ (t \geq 1)$	4	$+(8t+1)$, $-4t$, $+3t$, $-(6t+1)$
G	$n = 8t+7\ (t \geq 0)$	6	$+(8t+7)$, $+(t+1)$, $+(t+1)$, $-(2t+3)$, $-(2t+3)$, $-(2\,(t+1))$
H	$n = 24t+11\ (t \geq 0)$	6	$+(24t+11)$, $+24$, $-(2\,(25t+12))$, $+(25t+12)$, $-(25\,(2t+1))$, -4
I	$n = 24t+19\ (t \geq 0)$	6	$+1$, $+(24t+19)$, -3, $-(3\,(4t+3))$, $+(4t+3)$, $-(6t+5)$

Table 1: The nine path identities carrying n to $n+1$. Every integer $n \geq 2$ except $n = 3$ lies in exactly one class (Lemma 3.2).

Lemma 3.1 (Certified identities give paths)**.** *Fix a class $n = Mt + r$ with limit t_0 and a list of steps $(\varepsilon_i, N_i(t))_{i=1}^{L}$. Suppose polynomials $s_0(t) = Mt + r,\ s_1(t), \ldots, s_L(t) = Mt + r + 1$ and $k_1(t), \ldots, k_L(t)$ satisfy, for each i,*

$$\textit{forward: } s_{i-1} = k_i N_i, \quad s_i = k_i(N_i+1)^2;$$
$$\textit{inverse: } s_{i-1} = k_i(N_i+1)^2, \quad s_i = k_i N_i,$$

and that all $N_i(t)$ and $k_i(t)$ are integers ≥ 1 for every integer $t \geq t_0$. Then for every such t the sequence $s_0(t), \ldots, s_L(t)$ is a walk in G from n to $n+1$ of length L.

Proof. Each consecutive pair is, by the displayed relations, an edge $\{kN, k(N+1)^2\}$ with integers $k, N \geq 1$. □

A worked identity (class A). Let $n = 4t+1$ with $t \geq 1$. The walk is

$$4t+1 \xrightarrow{+(4t+1)} 4(2t+1)^2 \xrightarrow{-2t} 8t \xrightarrow{+t} 8(t+1)^2 \xrightarrow{-(2t+1)} 2(2t+1) = n+1.$$

Indeed: (1) $4t+1 = kN$ with $k=1$, $N = 4t+1$, giving $(4t+2)^2 = 4(2t+1)^2$. (2) $4(2t+1)^2 = k(N+1)^2$ with $N = 2t$, $k = 4$, giving $kN = 8t$. (3) $8t = kN$ with $N = t$, $k = 8$, giving $8(t+1)^2$. (4) $8(t+1)^2 = k(N+1)^2$ with $N = 2t+1$ and $k = 2$ (because $(N+1)^2 = 4(t+1)^2$), giving $kN = 2(2t+1) = n+1$. All labels and cofactors are positive integers for $t \geq 1$.

A second worked identity (class G). Let $n = 8t+7$, so $n+1 = 8(t+1)$. The walk is

$$\begin{aligned} 8t+7 &\xrightarrow{+(8t+7)} 64(t+1)^2 \xrightarrow{+(t+1)} 64(t+1)(t+2)^2 \xrightarrow{+(t+1)} 64(t+2)^4 \\ &\xrightarrow{-(2t+3)} 16(t+2)^2(2t+3) \xrightarrow{-(2t+3)} 4(2t+3)^2 \xrightarrow{-2(t+1)} 8(t+1) = n+1. \end{aligned}$$

The steps have labels and cofactors (N, k) equal to: $(8t+7,\ 1)$; $(t+1,\ 64(t+1))$; $(t+1,\ 64(t+2)^2)$; $(2t+3,\ 16(t+2)^2)$ [using $(N+1)^2 = 4(t+2)^2$]; $(2t+3,\ 4(2t+3))$ [same square]; and $(2t+2,\ 4)$ [using $(N+1)^2 = (2t+3)^2$]. In each forward step the state equals kN and the next state $k(N+1)^2$; in each inverse step the state equals $k(N+1)^2$ and the next state kN. All of these are positive integers for every $t \geq 0$.

Lemma 3.2 (Coverage)**.** *Every integer $n \geq 1$ is either 1 or 3, or lies in exactly one class: $n \equiv 1 \pmod 4$ is class A; $n \equiv 0, 2, 4, 6 \pmod 8$ are classes B, C, D, E; $n \equiv 7 \pmod 8$ is class G; and $n \equiv 3 \pmod 8$ is class F, H or I according as $n \equiv 3, 11, 19 \pmod{24}$. For $n = 1$ and $n = 3$ the explicit walks $1, 4, 18, 81, 8, 2$ and $3, 16, 4$ have lengths 5 and 2.*

Proof. The residues listed partition the integers; classes A, B and F require $t \geq 1$, which excludes only $n = 1$ (class A at $t = 0$) and $n = 3$ (class F at $t = 0$). The walk $1, 4, 18, 81, 8, 2$ uses the edges $\{1\cdot 1,\ 1\cdot 2^2\}$, $\{2\cdot 2,\ 2\cdot 3^2\}$, $\{9\cdot 2,\ 9\cdot 3^2\}$, $\{1\cdot 8,\ 1\cdot 9^2\}$ (traversed backwards) and $\{2\cdot 1,\ 2\cdot 2^2\}$ (backwards); the walk $3, 16, 4$ uses $\{1\cdot 3,\ 1\cdot 4^2\}$ and $\{4\cdot 1,\ 4\cdot 2^2\}$ (backwards). □

Proof of the upper bound in Theorem 1.1. By Lemma 3.2 it suffices to certify the nine identities. Appendix A gives, for each identity and each step, the label, the cofactor $k_i(t)$, and the states $s_{i-1}(t)$, $s_i(t)$ as explicit polynomials; each row is checked by one multiplication. All labels and cofactors are integer-valued polynomials with nonnegative coefficients and value ≥ 1 at $t = t_0$, so they are integers ≥ 1 for all $t \geq t_0$. Lemma 3.1 then gives walks of length at most 6. Together with Lemma 2.3 this proves $2 \leq \mathrm{d}(n, n+1) \leq 6$. □

Remark 3.3 (Refinement matters)**.** Using only residues modulo 8, the search found for the class $n \equiv 3 \pmod 8$ a uniform identity of length 7 and none shorter within its vocabulary of linear and quadratic label forms. Splitting it modulo 24 yields identities F, H, I of lengths 4, 6, 6. The constant $C_* = 6$ therefore depends on distinguishing n modulo 3 in this class.

4 Sharpness: the lower bound at $n = 163$

Lemma 4.1 (Meeting in the middle)**.** *If $\mathrm{d}(a, b) = t$ then $B_{\lceil t/2 \rceil}(a) \cap B_{\lfloor t/2 \rfloor}(b) \neq \emptyset$, where $B_r(x)$ is the set of vertices within distance r of x. In particular, $B_3(a) \cap B_2(b) = \emptyset$ implies $\mathrm{d}(a, b) \geq 6$.*

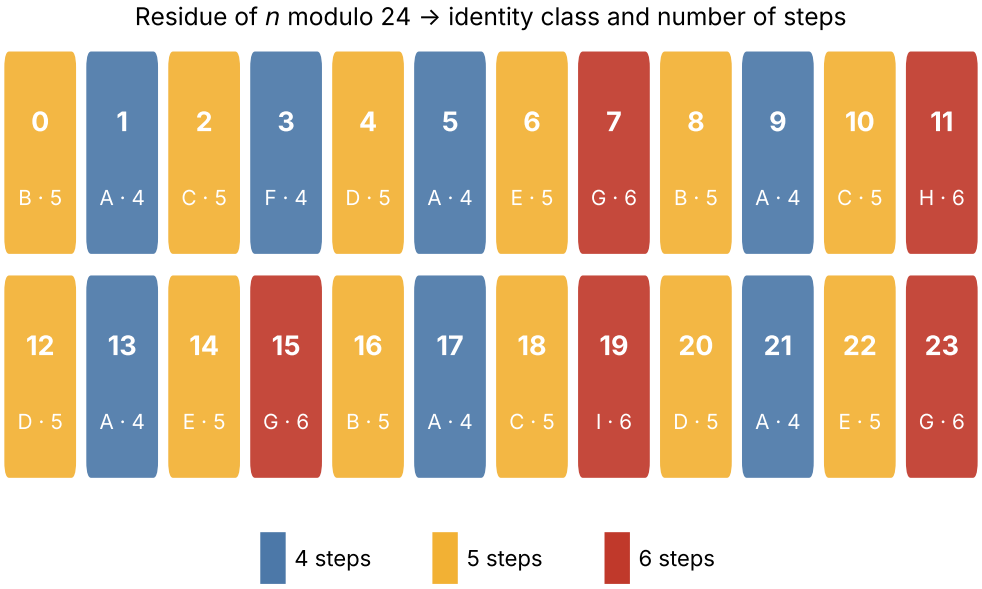


Figure 1: Residue classes modulo 24 and the identity (letter) assigned to each, with its number of steps. Only residues $7, 11, 15, 19, 23$ require six steps; classes F and A need four.

Proof. The vertex at position $\lceil t/2 \rceil$ along a shortest path lies in both balls. If $t \leq 5$ then $\lceil t/2 \rceil \leq 3$ and $\lfloor t/2 \rfloor \leq 2$, and balls grow with the radius. □

Theorem 4.2 (Lower bound)**.** $\mathrm{d}(163, 164) = 6$.

Proof (exhaustive computation). Balls are finite and computable by local finiteness. Enumerating them without any size cap gives $|B_3(163)| = 874$ and $|B_2(164)| = 123$, and $B_3(163) \cap B_2(164) = \emptyset$; by Lemma 4.1, $\mathrm{d}(163, 164) \geq 6$. Enumerating $B_3(164)$ (5599 vertices) gives 19 common vertices with $B_3(163)$, so a walk of length 6 exists (and identity I produces one explicitly: Figure 2). The computation was performed by two independent programs (one factoring neighbours with Pollard's method, one using only trial division); their ball sizes and intersections agree (check V8). □

The same computation gives $\mathrm{d}(211, 212) = \mathrm{d}(283, 284) = 6$; the first values with $\mathrm{d}(n, n+1) = 6$ are $163, 211, 283, 331, 523, 547, 571, 691$.

Corollary 4.3 (Linear cumulative cost)**.** $\sum_{n=1}^{k} \mathrm{d}(n, n+1) \leq 6k$ *for every* $k \geq 1$*; a walk through* $1, 2, \ldots, k+1$ *in order exists with at most* $6k$ *steps.*

5 Exact distances in the computed range

Using the exact search for every $n \leq 13500$ we obtain the distribution of Table 2; the mean is 4.43. The value 6 occurs 262 times in this range, and only at residues $11, 19$ modulo 24. This is consistent with Section 3: a value 6 can only occur where the assigned identity has length 6.

$d(n, n+1)$	2	3	4	5	6
count	2	23	7875	5338	262

Table 2: Exact $\mathrm{d}(n, n+1)$ for $1 \leq n \leq 13500$ (data file `successor_distance_table.csv`).

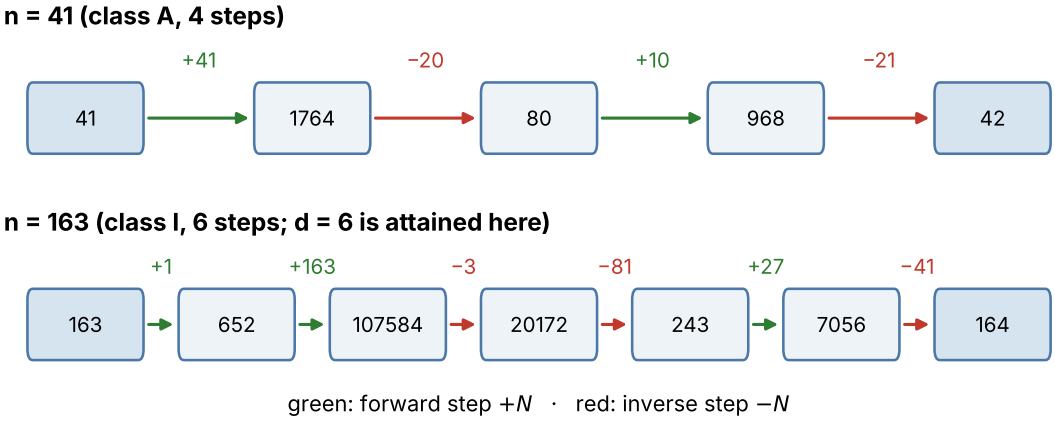


Figure 2: Two concrete walks produced by the identities: $n = 41$ by identity A (four steps) and $n = 163$ by identity I (six steps, optimal).

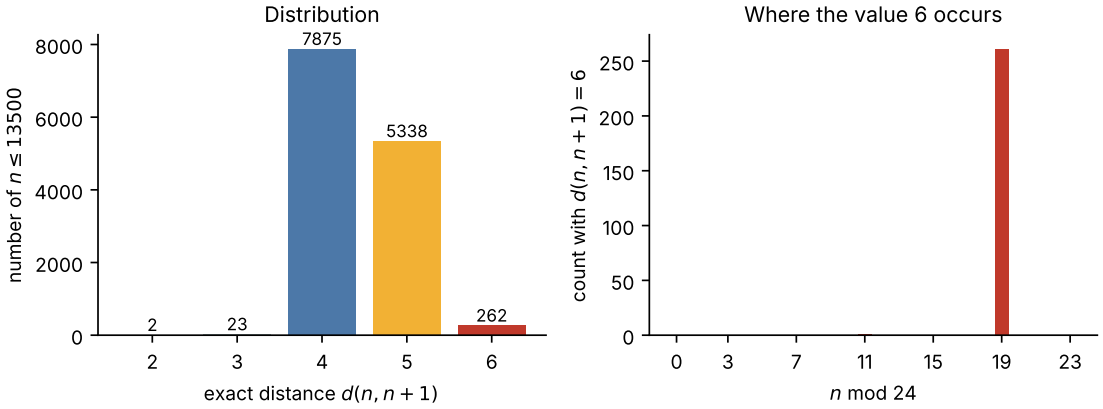


Figure 3: Left: distribution of exact distances. Right: residues modulo 24 at which $\mathrm{d}(n, n+1) = 6$ occurs within the computed range.

Remark 5.1. Computation alone cannot show that $\mathrm{d}(n, n+1) = 6$ occurs infinitely often; it is observed 262 times below 13500. The theorem only asserts that the supremum equals 6.

6 Status of the results

Statement	Status
Upper bound $\mathrm{d}(n, n+1) \leq 6$	Proved: Lemmas 3.1, 3.2 and the certificates; computer-algebra verified; classes A and G checked by hand in the text
Lower bound $\mathrm{d}(n, n+1) \geq 2$	Proved (Lemma 2.3)
Sharpness $\mathrm{d}(163, 164) = 6$	Exhaustive computation, two independent programs; not a hand proof
Occurrence of the value 6 for infinitely many n	Not established
Connectivity of G	Context only (preceding note [2]); constructive paths re-checked for $n \leq 1500$ (V5)
Proof-assistant certification	Not performed
Originality relative to prior literature	Not reviewed

Table 3 lists the finite computations. The symbolic check V2 is an exact computer-algebra statement (`sympy` [5]); the others are finite computations over the stated ranges and support,

but do not replace, the proofs above.

check	what it tests	cases	result
V1	closed-form edge test equals the definition	40000	pass
V2	exact symbolic certificates for the nine identities	9	pass
V3	coverage of all residues modulo 24; special paths for $n = 1, 3$	97	pass
V4	plain-integer replay of every identity with edge test	400000	pass
V5	constructive connectivity paths (context)	1500	pass
V6	exact distance table, consistency with identity lengths	13500	pass
V7	independent engine reproduces distances	62	pass
V8	lower-bound certificate at $n = 163$, two engines	3	pass
V9	trivial lower bound $d(n, n+1) \geq 2$	2	pass

Table 3: Verification checks (455173 cases in total, deterministic exact arithmetic, report `u1_bnd_verification_report.json`).

7 Consequences and observations

Waypoint cost. By Corollary 4.3, the sequence $1, 2, 3, \ldots$ has bounded *amortized* waypoint cost in G: at most six legal transitions per successor, whatever the magnitude of n; in the computed range the mean is 4.43.

What this does and does not say. The identities are uniform programs of fixed length with n entering only through a class parameter; to apply one it suffices to know which of finitely many residue classes contains n. This is a statement about the geometry of G. It is *not* a selector: it does not show that a rule computing each next label from the current state alone, without being told the target, reproduces the integers. That question is separate.

8 Open questions

1. Does $\mathrm{d}(n, n+1) = 6$ occur for infinitely many n? Is there a density statement in classes H and I?

2. Which residue classes modulo larger moduli admit uniform identities shorter than those above, and which are exact?

3. For which rational functions f is $\sup_n \mathrm{d}(n, f(n))$ finite (for example $f(n) = 2n,\ n+2,\ n^2$)? The search method used here (in `code/search`) finds candidate identities for such pairs.

4. The growth of $\mathrm{d}(1, n)$: the coarse bound known from the preceding note is linear; the true growth rate is unknown.

5. Can a fixed state-computable rule select the labels, so that the successor transition is generated rather than supplied?

A Complete step certificates

Each row records one step: type, label N, cofactor k, and the states before and after. A forward row satisfies (before)$= kN$ and (after)$= k(N+1)^2$; an inverse row satisfies (before)$= k(N+1)^2$ and (after)$= kN$. These tables are generated from the symbolic certificates (check V2).

Identity A: $n = 4t + 1$, $t \geq 1$ **(4 steps).**

step	type	N	cofactor k	state before	state after
1	forward	$4t+1$	1	$4t+1$	$4(2t+1)^2$
2	inverse	$2t$	4	$4(2t+1)^2$	$8t$
3	forward	t	8	$8t$	$8(t+1)^2$
4	inverse	$2t+1$	2	$8(t+1)^2$	$2(2t+1)$

Identity B: $n = 8t$, $t \geq 1$ **(5 steps).**

step	type	N	cofactor k	state before	state after
1	forward	2	$4t$	$8t$	$36t$
2	forward	2	$18t$	$36t$	$162t$
3	inverse	8	$2t$	$162t$	$16t$
4	forward	$4t$	4	$16t$	$4(4t+1)^2$
5	inverse	$8t+1$	1	$4(4t+1)^2$	$8t+1$

Identity C: $n = 8t + 2$, $t \geq 0$ **(5 steps).**

step	type	N	cofactor k	state before	state after
1	forward	2	$4t+1$	$2(4t+1)$	$9(4t+1)$
2	forward	$4t+1$	9	$9(4t+1)$	$36(2t+1)^2$
3	forward	2	$18(2t+1)^2$	$36(2t+1)^2$	$162(2t+1)^2$
4	inverse	8	$2(2t+1)^2$	$162(2t+1)^2$	$16(2t+1)^2$
5	inverse	$8t+3$	1	$16(2t+1)^2$	$8t+3$

Identity D: $n = 8t + 4$, $t \geq 0$ **(5 steps).**

step	type	N	cofactor k	state before	state after
1	forward	2	$2(2t+1)$	$4(2t+1)$	$18(2t+1)$
2	forward	2	$9(2t+1)$	$18(2t+1)$	$81(2t+1)$
3	inverse	8	$2t+1$	$81(2t+1)$	$8(2t+1)$
4	forward	$2(2t+1)$	4	$8(2t+1)$	$4(4t+3)^2$
5	inverse	$8t+5$	1	$4(4t+3)^2$	$8t+5$

Identity E: $n = 8t + 6$, $t \geq 0$ **(5 steps).**

step	type	N	cofactor k	state before	state after
1	forward	2	$4t+3$	$2(4t+3)$	$9(4t+3)$
2	forward	$4t+3$	9	$9(4t+3)$	$144(t+1)^2$
3	forward	2	$72(t+1)^2$	$144(t+1)^2$	$648(t+1)^2$
4	inverse	8	$8(t+1)^2$	$648(t+1)^2$	$64(t+1)^2$
5	inverse	$8t+7$	1	$64(t+1)^2$	$8t+7$

Identity F: $n = 24t + 3$, $t \geq 1$ **(4 steps).**

step	type	N	cofactor k	state before	state after
1	forward	$8t+1$	3	$3(8t+1)$	$12(4t+1)^2$
2	inverse	$4t$	12	$12(4t+1)^2$	$48t$
3	forward	$3t$	16	$48t$	$16(3t+1)^2$
4	inverse	$6t+1$	4	$16(3t+1)^2$	$4(6t+1)$

Identity G: $n = 8t + 7$, $t \geq 0$ **(6 steps).**

step	type	N	cofactor k	state before	state after
1	forward	$8t + 7$	1	$8t + 7$	$64\,(t + 1)^2$
2	forward	$t + 1$	$64\,(t + 1)$	$64\,(t + 1)^2$	$64\,(t + 1)\,(t + 2)^2$
3	forward	$t + 1$	$64\,(t + 2)^2$	$64\,(t + 1)\,(t + 2)^2$	$64\,(t + 2)^4$
4	inverse	$2t + 3$	$16\,(t + 2)^2$	$64\,(t + 2)^4$	$16\,(t + 2)^2\,(2t + 3)$
5	inverse	$2t + 3$	$4\,(2t + 3)$	$16\,(t + 2)^2\,(2t + 3)$	$4\,(2t + 3)^2$
6	inverse	$2\,(t + 1)$	4	$4\,(2t + 3)^2$	$8\,(t + 1)$

Identity H: $n = 24t + 11$, $t \geq 0$ **(6 steps).**

step	type	N	cofactor k	state before	state after
1	forward	$24t + 11$	1	$24t + 11$	$144\,(2t + 1)^2$
2	forward	24	$6\,(2t + 1)^2$	$144\,(2t + 1)^2$	$3750\,(2t + 1)^2$
3	inverse	$2\,(25t + 12)$	6	$3750\,(2t + 1)^2$	$12\,(25t + 12)$
4	forward	$25t + 12$	12	$12\,(25t + 12)$	$12\,(25t + 13)^2$
5	inverse	$25\,(2t + 1)$	3	$12\,(25t + 13)^2$	$75\,(2t + 1)$
6	inverse	4	$3\,(2t + 1)$	$75\,(2t + 1)$	$12\,(2t + 1)$

Identity I: $n = 24t + 19$, $t \geq 0$ **(6 steps).**

step	type	N	cofactor k	state before	state after
1	forward	1	$24t + 19$	$24t + 19$	$4\,(24t + 19)$
2	forward	$24t + 19$	4	$4\,(24t + 19)$	$64\,(6t + 5)^2$
3	inverse	3	$4\,(6t + 5)^2$	$64\,(6t + 5)^2$	$12\,(6t + 5)^2$
4	inverse	$3\,(4t + 3)$	3	$12\,(6t + 5)^2$	$9\,(4t + 3)$
5	forward	$4t + 3$	9	$9\,(4t + 3)$	$144\,(t + 1)^2$
6	inverse	$6t + 5$	4	$144\,(t + 1)^2$	$4\,(6t + 5)$

B Reproducibility

The package contains the verifier (`code/verify_u1_bnd.py`, SHA-256 prefix bd313fce5b368633), the library with the engines and identities, the search tool used to discover the identities, the exact-distance table, the machine-readable identity record and the verification report. Run `python3 verify_u1_bnd.py` inside `code/` (requires Python 3 and `sympy`); the report is deterministic.

References

[1] UNNS Research Collective, *UNNS-U0: Arithmetic reachability and orbit admissibility*, research note U0-001, 8 November 2026 (local release; not part of the frozen repository).

[2] UNNS Research Collective, *UNNS-U1: Reversible arithmetic connectivity*, research note, 8 November 2026 (provisional baseline; local release).

[3] UNNS Research Collective, *UNNS Generative Information Geometry collection*, frozen release, repository `ukbbi/UNNS`, commit `a115e4bc6b8ee734e6146b3b17a6b121e65d31c3`: https://github.com/ukbbi/UNNS/tree/a115e4bc6b8ee734e6146b3b17a6b121e65d31c3/projects/UNNS_Generative_Information_Geometry.

[4] UNNS project repository, pinned at the same commit: https://github.com/ukbbi/UNNS/tree/a115e4bc6b8ee734e6146b3b17a6b121e65d31c3.

[5] A. Meurer et al., SymPy: symbolic computing in Python, *PeerJ Computer Science* **3** (2017), e103, https://doi.org/10.7717/peerj-cs.103.